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RESEARCH ARTICLE

Investigation of the contribution of total creatine to the CEST *Z*-spectrum of brain using a knockout mouse model

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The current study aims to assign and estimate the total creatine (tCr) signal contribution to the *Z*-spectrum in mouse brain at 11.7 T. Creatine (Cr), phosphocreatine (PCr) and protein phantoms were used to confirm the presence of a guanidinium resonance at this field strength. Wild-type (WT) and knockout mice with guanidinoacetate *N*-methyltransferase deficiency (GAMT−/−), which have low Cr and PCr concentrations in the brain, were used to assign the tCr contribution to the *Z*-spectrum. To estimate the total guanidinium concentrations, two pools for the *Z*-spectrum around 2 ppm were assumed: (i) a Lorentzian function representing the guanidinium chemical exchange saturation transfer (CEST) at 1.95 ppm in the 11.7-T *Z*-spectrum; and (ii) a background signal that can be fitted by a polynomial function. Comparison between the WT and GAMT−/− mice provided strong evidence for three types of contribution to the peak in the *Z*-spectrum at 1.95 ppm, namely proteins, Cr and PCr, the latter fitted as tCr. A ratio of 20 ± 7% (protein) and 80 ± 7% tCr was found in brain at 2 µT and 2 s saturation. Based on phantom experiments, the tCr peak was estimated to consist of about 83 ± 5% Cr and 17 ± 5% PCr. Maps for tCr of mouse brain were generated based on the peak at 1.95 ppm after concentration calibration with *in vivo* magnetic resonance spectroscopy.

KEYWORDS
chemical exchange saturation transfer (CEST), creatine, guanidinoacetate N-methyltransferase deficiency (GAMT−/−), Lorentzian line-shape fitting, magnetic resonance spectroscopy (MRS), magnetization transfer, phosphate creatine

1 | INTRODUCTION

Chemical exchange saturation transfer (CEST) magnetic resonance imaging (MRI), as a novel molecular imaging technique, has received great attention in recent years because of its potential to detect low concentration proteins and metabolites *in vivo*.[1-3] The technique has been successfully applied to detect pathological chemical changes in many diseases, such as neurological disorders in human brain,[4] animal models of cerebral ischemia,[5-7] and cancer in both animal models and humans.[8-14] CEST MRI detects the exchangeable protons present in proteins, and metabolites *in vivo*, whose chemical shifts are mainly in the offset range of 0 to 5 ppm with respect to the water resonance, such as amide protons around 3.6 ppm,

Abbreviations used: Ala, alanine; Asp, aspartate; CERT, chemical exchange rotation transfer; CEST, chemical exchange saturation transfer; CW, continuous wave; DS, direct water saturation; FLEX, frequency-labeled exchange transfer; FWHM, full width at half-maximum; FOV, field of view; GABA, γ-aminobutyric acid; GAMT−/−, guanidinoacetate *N*-methyltransferase deficiency/deficient; Gln, glutamine; Glu, glutamate; Glx, Glu/Gln; GPC, glycerophosphorylcholine; GSH, glutathione; Ins, myo-inositol; Lac, lactate; MRI, magnetic resonance imaging; MRS, magnetic resonance spectroscopy; MTC, magnetization transfer contrast; NAA, *N*-acetylaspartate; NMR, nuclear magnetic resonance; OVS, outer volume suppression; PBS, phosphate-buffered saline; PCho, phosphorylcholine; PCr, phosphocreatine; PE, phosphorylethanolamine; RARE, rapid acquisition with relaxation enhancement; RF, radiofrequency; rNOE, relayed nuclear Overhauser enhancement; ROI, region of interest; Ser, serine; Si, scyllo-inositol; ST, saturation transfer; STEAM, stimulated echo acquisition mode; Tau, taurine; TSE, turbo spin echo; VAPOR, variable power RF pulses and optimized relaxation delays; VDMP, variable delay multi-pulse; VOI, volume of interest; WASSR, water saturation shift referencing; WT, wild-type

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amine groups around 1.5–3 ppm and hydroxyl groups around 1 ppm.[15] The abundant exchangeable protons in tissues cause significant overlap in the *Z*-spectrum between 0 and 5 ppm. The situation worsens for the fast-exchanging (exchange rate k > 1 kHz) amine and hydroxyl protons, the lines of which broaden and may even coalesce with the water proton resonance at lower fields. In the chemical shift offset range between 0 and 5 ppm, it has been found that non-exchangeable aromatic protons also contribute to the *Z*-spectra through a relayed nuclear Overhauser enhancement (rNOE) CEST effect.[1,16] Together, these factors cause a lack of specificity in the *Z*-spectrum of endogenous exchanging protons. To date, efforts have been made towards the separation of the amide and aliphatic protons from other contributions by exploiting foremost the abundance of proteins in tissues and the slower exchange rates of amides and rNOEs.[5,6,8,17,18] There are several CEST techniques that can detect amide protons and aliphatic protons with reasonable specificity, such as frequency-labeled exchange transfer (FLEX),[19] the three-point fitting method[20,21] and the variable delay multi-pulse (VDMP) technique.[22,23]

Despite significant efforts, the assignment and selective detection of endogenous metabolites in tissue with CEST are still challenging. A rough selectivity can sometimes be achieved by tuning the labeling radiofrequency (RF) power, frequency and duration[24-27] for particular metabolites, such as glycogen,[28] glycosaminoglycan,[29] glutamate,[26,30] creatine[27,31] and lactate.[32] However, contamination from lipids, proteins, semisolid macromolecules and other metabolites complicates this simplified approach. The chemical exchange rotation transfer (CERT) technique is capable of measuring the creatine (Cr) CEST signal in phantoms at 9.4 T.[33-35] Recently, a CEST method that fits the *in vivo Z*-spectrum using a group of Lorentzian line-shapes[12,13,36] has been proposed, which achieves improved selectivity by partially fitting out the semi-solid tissue component and direct water saturation (DS),[7,12] and avoiding asymmetry analysis-based contamination with aliphatic CEST signals.

The CEST *Z*-spectra recorded in the brain and muscles at low magnetic fields (3- and 7-T MRI) and typical B_1 values (1 μT or more) are broad and featureless, but estimated maps of changes in Cr concentration in tissue have been published using asymmetry analysis on the CEST *Z*-spectra.[27] However, high-field *Z*-spectra of the healthy brain show a well-defined peak at 1.95 ppm on top of a broad background,[12,13] which offers an opportunity to estimate the contributions from total creatine (tCr) in the CEST *Z*-spectrum. The broad background is considered to be a combination of DS, semi-solid magnetization transfer contrast (MTC) and several metabolites with fast-exchanging amine protons, such as glutamate (Glu), glutamine (Gln), taurine (Tau) and γ-aminobutyric acid (GABA), whereas the sharper peak in this frequency range has been thought to be mainly from the guanidinium protons of Cr,[7,12] considering that the exchange rate of these protons is in the intermediate range *in vivo*.[37,38] One recent study on brain tumors concluded that a significant part of the CEST peak at 1.95 ppm originates from Cr.[12] However, it is well known that the metabolites and macromolecules in tumors are significantly different from those in healthy brain tissue,[39-41] which complicates the CEST signal detected at 1.95 ppm. Apart from comparison studies with phantoms or *ex vivo* tissue homogenates,[42] assignment of Cr has not been validated *in vivo* and other contributions to the peak may be present.

In this article, the contribution to the well-defined CEST signal at 1.95 ppm in the *Z*-spectrum was investigated by comparing wild-type (WT) mice with knockout mice having guanidinoacetate *N*-methyltransferase deficiency (GAMT−/−), which are known to have very low Cr and phosphocreatine (PCr) concentrations.[43-47] This allowed the total Cr contribution (tCr = Cr + PCr) to be separated out. Assuming that the residual stems from guanidinium in mobile proteins (based on phantom data), and using a polynomial background removal method[48] combined with *in vivo* magnetic resonance spectroscopy (MRS) calibration, we derived a quantitative estimation of tCr concentration.

2 | METHODS

2.1 | MRI experiments

All MRI experiments were performed on a horizontal-bore, 11.7-T Bruker Biospec system (Bruker, Ettlingen, Germany). A 23-mm volume transceiver coil was used for phantom experiments, and MR images on animals were acquired using a 72-mm quadrature volume resonator as a transmitter and a four-element (2 × 2) phased array coil as a receiver. The CEST experiments were performed using continuous wave (CW) CEST saturation pulses with a length of 2 s. The saturation offset sweeps were from 1 to 5 ppm at 0.1-ppm increments for both phantoms and mouse brain. In the mouse brain study, a 0.05-ppm increment between 1.6 and 2.4 ppm was added, which leads to a scan time of 5 min per *Z*-spectrum. MR images were acquired using a turbo spin echo (TSE) sequence with TR/TE = 5 s/3.7 ms, TSE factor = 23, slice thickness of 2 mm, matrix size of 64 × 64 and resolution of 0.25 × 0.25 mm^2. The B_0 field over the mouse brain was adjusted using field mapping and second-order shimming. A water saturation shift referencing (WASSR) method was applied to check the B_0 map after shimming.[49] For the phantom studies, the *Z*-spectrum was adjusted based on the position of the guanidinium peak with respect to the B_0 map determined by WASSR. The B_0 map on mouse brain is not necessary as the concentration was obtained by the magnitude of the peak at 1.95 ppm. The R_1 relaxation times of the mouse brain and phantoms were measured using variable TR (TR = 0.5, 1, 1.5, 2, 3.5, 5 and 8 s) with the aforementioned imaging parameters. The *in vivo* MRS experiments were acquired on a voxel of 2 × 2 × 2 mm^3 using a stimulated echo acquisition mode (STEAM) sequence (TE = 3 ms; TM = 10 ms; TR = 2.5 s) with outer volume suppression (OVS). Unsuppressed water signal was acquired prior to each MRS experiment with the same parameters as the MRS experiment, except that the number of averages was one. First- and second-order shims were adjusted for each volume of interest (VOI) using a field mapping method. The water signal was suppressed by the variable power RF pulses and optimized relaxation delays (VAPOR) method.[50] Phantoms with Cr, GABA, Glu, PCr, eggwhite and hair conditioner (Suave) were selected to represent metabolites found in brain tissue with exchanging protons around 1.95 ppm, as well as mobile protein and semisolid MTC pools,[51-53] respectively. Except for the hair conditioner and eggwhite, phantoms were prepared in phosphate-buffered saline (PBS), titrated to pH 7.3 (typical brain cellular pH) and transferred to 5-mm nuclear magnetic resonance (NMR) tubes. The phantoms were maintained at 37°C during MRI experiments by an air heater.

2.2 | Animal studies

The institutional animal care and use committee approved the study. Three adult female BALB/c mice (6 months) as WT mice and three GAMT−/− mice (8–12 weeks) were used for the experiments. ^{1}H MRS spectra of both the brain and muscle of GAMT−/− mice have been reported to show greatly reduced levels of tCr relative to WT animals.[43-47] Although the age is different, the GAMT−/− mice were used as a model with reduced tCr levels, for which this age group sufficed. The experiments on the brain of WT mice were performed on the midbrain and cerebellum. These two brain regions were chosen because of the significantly different tCr concentrations (12.5–13.5 μmol/g for cerebellum and 9–11 μmol/g for midbrain) reported previously.[54,55] All animals were induced using 2% vaporized inhaled isoflurane, followed by 1.5% isoflurane during the MRI scan.

2.3 | CEST and MRS quantification

At 1.95 ppm, the water proton Z-spectrum shows contributions from DS and a saturation transfer (ST) signal from CEST effects from guanidinium protons, other exchanging protons and semisolid macromolecules. These contributions add up non-linearly and, to make matters worse, their relative contribution changes with B_1 strength and length. In the current study, we employed inverse metric evaluation[20,56-59] to extract the contribution from guanidinium proton CEST signals. In the framework of this method, the CEST effect under CW saturation can be treated as a relaxation process similar to $R_{1\rho}$ experiments.[24,56] Then, the normalized saturation signal Z (i.e. the water saturation signal S normalized by the signal without saturation S_0) at each offset is given by[24,56,57]:

$$Z = (1-Z^{ss})e^{-R_{1\rho}t_{sat}} + Z^{ss} \tag{1}$$

$$Z^{ss} = \frac{cos^2\theta R_1}{R_{1\rho}} \tag{2}$$

where Z^{ss} is the steady-state Z-spectrum, R_1 is the longitudinal relaxation time of water, t_{sat} is the saturation time and $\theta = tan^{-1}\omega_1/\Delta$ is the tilt angle of the effective magnetization with respect to the z-axis as a result of RF saturation with a nutation frequency of ω_1 and an offset of Δ. $R_{1\rho}$ is the water relaxation time under RF saturation, which contains contributions from the effective water relaxation that accounts for the direct saturation effect in Z-spectra and an apparent relaxation term from all ST processes in the tissue.[57]

$$R_{1\rho} = R_{eff} + R_{ST} \tag{3}$$

where $R_{eff} = \cos^2\theta R_1 + \sin^2\theta R_2$ is the measured longitudinal relaxation time of water in the rotating frame without additional solution components. Equation 3 can be correlated with the observed Z-spectrum using Equations 1 and 2. When the saturation time is long enough to reach the steady-state condition, R_{ST} can be calculated from the steady-state CEST signal (Z^{ss}) using a simple analytical equation resulting from the combination of Equations 2 and 3:

$$R_{ST} = \frac{cos^2\theta R_1}{Z^{ss}} - R_{eff} \tag{4}$$

The steady-state condition is determined by $R_{1\rho}$, which is usually much faster than the water relaxation rate R_1 in the tissue because of the presence of a large MTC pool and the abundant exchanging protons. $R_{1\rho}$ of mouse brain was measured at 11.7 T, and was found to be 1.6 ± 0.1 s^{-1} for a saturation power of 2 μT at an offset of 1.95 ppm, which indicates that a saturation time of 2 s is sufficient for a steady-state condition in mouse brain. As a comparison, R_1 of mouse brain was determined to be 0.55 ± 0.03 s^{-1} at 11.7 T.

The steady-state Z-spectrum (Z^{ss}) is modulated by the water longitudinal relaxation time R_1, and is not linearly dependent on the concentration of exchanging protons.[20,56-59] These limitations can be addressed by using R_{ST}. Another great advantage of using R_{ST} is that it is a linear superposition for all ST components when multiple exchanging proton pools are present. For the two-pool situation used in the current study, i.e. a guanidinium pool and one background pool that includes all other contributing protons at that frequency in the Z-spectrum, R_{ST} is given by:

$$R_{ST} = R_{background} + R_{guan} \tag{5}$$

Importantly, and not fully intuitively, to calculate individual component apparent rate contributions, Equation 4 must be applied to each individual component to fulfill the conditions of the addition of inverse spectral intensities. Thus:

$$R_{background} = \frac{R_1 cos^2\theta}{Z^{ss}_{background}} - R_{eff} \tag{6}$$

$$R_{guan} = \frac{R_1 cos^2\theta}{Z_{guan}^{ss}} - R_{eff} \tag{7}$$

where $Z_{background}^{ss}$ and Z_{guan}^{ss} are the steady-state CEST signals when only a single background or guanidinium pool is present, respectively. By combining Equations 4–7, the guanidinium CEST signal can be extracted by fitting the steady-state guanidinium Z-spectrum Z_{guan}^{ss} using:

$$\left[\frac{1}{Z^{ss}} - \frac{R_{eff}}{cos^2\theta R_1}\right] = \left[\frac{1}{Z_{background}^{ss}} - \frac{R_{eff}}{cos^2\theta R_1}\right] + \left[\frac{1}{Z_{guan}^{ss}} - \frac{R_{eff}}{cos^2\theta R_1}\right] \tag{8}$$

Equation 8 can be easily extended to the situation with more than two exchanging proton pools. Because of the somewhat complicated arithmetics for spectral intensities, R_{guan} is a better option than Z_{guan}^{ss} for the quantification of CEST signals. Therefore, in the current study, R_{guan} is adopted for the quantification of the guanidinium protons by combining Equations 7 and 8.

An example of the observed steady-state Z-spectrum based on the assumption of two contributions, R_{guan} and $Z_{background}^{ss}$, is simulated in Figure 1. The observed guanidinium CEST signal $\Delta Z = \left(Z_{background}^{ss} - Z^{ss}\right)$ and the maximum R_{guan}, R_{guan}^{max}, are indicated in the figure. With the assumption that $\Delta Z \ll Z^{ss}$ and $cos^2\theta \approx 1$, the analytical equation of ΔZ can be derived from Equation 8, and is given by:

$$\Delta Z \approx Z^{ss} \cdot Z_{background}^{ss}\left(1 - Z_{guan}^{ss}\right) \tag{9}$$

When strong background ST effects ($Z_{background}^{ss}$) are present, which is the case for the *in vivo* CEST experiments, the observed guanidinium peak (ΔZ) will be scaled down significantly from the guanidinium Z-spectrum alone ($1 - Z_{guan}^{ss}$) (see Figure 1), which is called a spillover effect. The Z-spectra between 1 and 3 ppm were fitted by assuming two saturation components: a well-defined peak, assigned to guanidinium protons (R_{guan}), and the broad background signal including DS and ST effects from MTC, aromatic protons and other metabolites ($Z_{background}^{ss}$). These components were fitted using a Lorentzian and a fourth-order polynomial function,[48] respectively.

$$R_{guan} = R_{guan}^{max} \frac{(w/2)^2}{(w/2)^2 + (\Delta - \Delta_{guan})^2} \tag{10}$$

$$Z_{background}^{ss} = C0 + C1(\Delta - \Delta_{guan}) + C2(\Delta - \Delta_{guan})^2 + C3(\Delta - \Delta_{guan})^3 + C4(\Delta - \Delta_{guan})^4 \tag{11}$$

where w is the peak full width at half-maximum (FWHM) of the Lorentzian line-shape in ppm. R_{guan}^{max} is the true apparent relaxation rate contribution of the guanidinium protons. The pure guanidinium steady-state CEST signal Z_{guan}^{ss} can be calculated from R_{guan} using Equation 7. Δ_{guan} is the chemical shift of the guanidinium peak, which is around 1.95 ppm *in vivo*; the terms $C0$–$C4$ are the zero- to fourth-order polynomial coefficients. A lower order polynomial function (<4) was not able to fit the *in vivo* broad spectral background around 2 ppm with sufficient accuracy, as demonstrated by the fitting on a typical Z-spectrum on a mouse brain plotted in Figure 1B, and a higher order polynomial function may cause over-fitting.

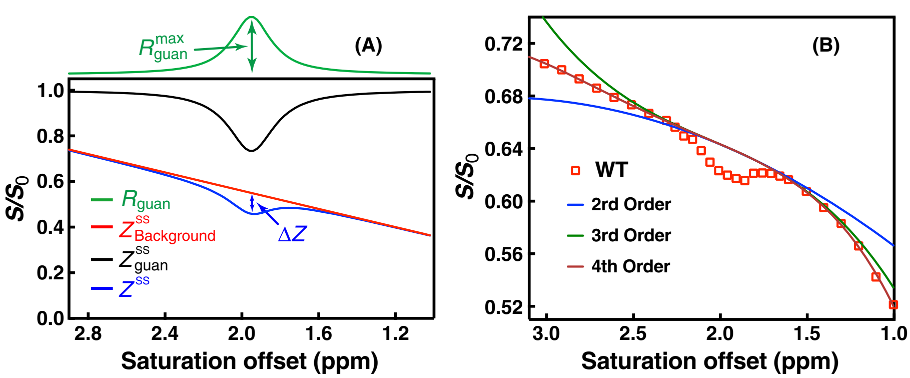

FIGURE 1 (A) Example of fitting of the Z-spectrum (blue line) consisting of two contributions (R_{guan} and $Z_{background}^{ss}$) simulated using equations 7, 8, 10 and 11. The corresponding Z_{guan}^{ss} calculated from R_{guan} using equation 7 with $R_1 = 0.5\ s^{-1}$ is also plotted (black line). In the fitting, $R_{guan}^{max} = 0.2\ s^{-1}$, $w = 0.250$ ppm and $\Delta_{guan} = 1.95$ ppm were used for the Lorentzian line-shape (R_{guan}), and $C0 = 0.55$ and $C1 = -0.2$ were applied for the broad background signal ($Z_{background}^{ss}$). Other polynomial terms were assumed to be zero in $Z_{background}^{ss}$. ΔZ is the observed pure guanidinium chemical exchange saturation transfer (CEST) signal. (B) One typical Z-spectrum recorded in a wild-type (WT) mouse with a saturation power of 1 μT and a saturation length of 2 s (open squares) is plotted together with the $Z_{background}^{ss}$ fitting using several polynomial functions with different orders (full lines) defined by equation 11

The fitting of the guanidinium peak was performed by two steps. First, the $Z_{background}$ component was fitted by excluding the CEST data points between 1.6 and 2.4 ppm and varying the parameters *C*0 – *C*4. Then, the *Z*-spectral data points between 1.6 and 2.4 ppm were fitted by varying the parameters *w*, Δ_{guan} and R_{guan}^{max} to superimpose the Z_{guan}^{ss} and $Z_{background}^{ss}$ components using Equations 7 and 8.

Brain tCr concentrations from the *in vivo* ^{1}H MRS spectra were estimated using LCModel[60,61] analysis. The unsuppressed water signal measured from the same VOI was used as an internal reference for the quantification (assuming 80% brain water content). The following 16 compounds were used for a basis set: alanine (Ala); aspartate (Asp); Cr; GABA; Glu; Gln; glutathione (GSH); glycerophosphorylcholine (GPC); phosphorylcholine (PCho); myo-inositol (Ins); lactate (Lac); *N*-acetylaspartate (NAA); phosphorylethanolamine (PE); scyllo-inositol (Si); serine (Ser); and Tau. The model spectrum of PCr was not included in the fitting and the resulting Cr concentration thus represents tCr. As a result of the overlap of the Cr and PCr signals in the proton spectrum, the concentration ratio between them is hard to determine by high-resolution *in vivo* MRS, even with a small voxel at high field. In the literature, an approximate ratio of 1: 1 has been reported for Cr/PCr,[54,62-65], which could be used to extract the Cr or PCr concentration in mouse brain. The metabolite concentrations were obtained from the ratio of the metabolite peak area from LCModel to that of the unsuppressed water signal.

3 | RESULTS

3.1 | Phantom experiments

Figure 2A shows phantom *Z*-spectra measured for several abundant mouse brain metabolites with a CEST signal around 1.95 ppm using a CW-CEST sequence with a saturation field of 2 μT (t_{sat} = 2 s). The results are similar to previously reported *Z*-spectra.[25,31] At the resonance of 1.95 ppm, Glu, GABA, Cr, MTC (hair conditioner) and mobile protein (eggwhite) give strong CEST signals. However, most of the metabolites have broad line-shapes, except for Cr, PCr and protein guanidinium protons, which all show a well-defined resonance at 1.95 ppm. PCr has an additional CEST signal around 2.6 ppm. The R_{guan}^{max} values for tCr and PCr CEST at 1.95 ppm were 0.102 ± 0.009 s^{-1} and 0.020 ± 0.003 s^{-1} for 10mM concentration, respectively, which translate to 83 ± 5% and 17 ± 5% contributions to tCr under these saturation conditions and a 1: 1 Cr/PCr concentration ratio. The R_{guan}^{max} values were calculated using Equation 1 with the measured water longitudinal relaxation time $R_1 = 0.35 \pm 0.05$ s^{-1}. In the phantoms, the CEST signal cannot reach steady-state with a saturation time of 2 s. Therefore, the non-steady-state equation (Equation 1) was used for the calculation of R_{guan}^{max}. The concentration dependence of the R_{guan}^{max} signal is illustrated in Figure 2B. For these concentration studies *in vitro*, the Cr CEST effect (R_{guan}^{max}) is linearly dependent on the concentration up to about 20mM (Figure 2C).

3.2 | MR of mouse brain

In vivo ^{1}H MR spectra of WT mouse brain (cerebellar and midbrain regions) and the midbrain of a GAMT−/− mouse are shown in Figure 3. The high-resolution T_2-weighted images acquired with a rapid acquisition with relaxation enhancement (RARE) sequence are also shown for anatomical guidance for reproducible placement of a VOI. The tCr signal in the midbrain of the GAMT−/− mouse shows striking differences with respect to the spectrum of the WT mouse, being significantly reduced at both 3 ppm ($p < 0.001$) and 3.9 ppm ($p < 0.001$). It should be noted that the residual peak at 3.0 ppm is not attributed to tCr, but can be assigned to mobile macromolecules, resulting from the consideration of these in the LCModel fitting. However, LCModel-based concentrations of GABA and Glu/Gln (Glx) at this frequency did not show a significant difference ($p = 0.93$ for Glx, $p = 0.32$ for GABA) (Table 1).

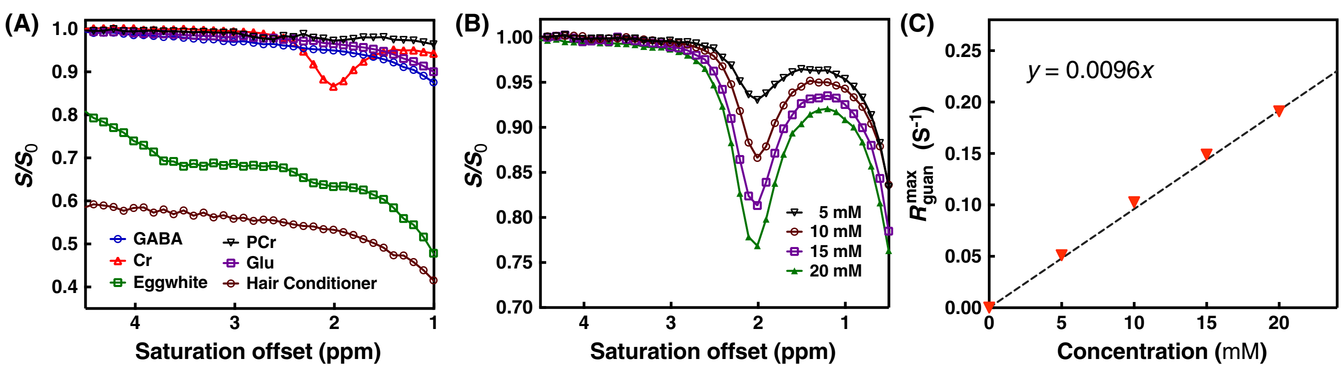

FIGURE 2 *Z*-spectra of solutions of glutamate (Glu) (10mM), phosphocreatine (PCr) (10mM), creatine (Cr) (10mM) and γ-aminobutyric acid (GABA) (10mM), hair conditioner [magnetization transfer contrast (MTC) phantom] and eggwhite (protein phantom) (A) and Cr solutions with different concentrations (5, 10, 15 and 20mM) (B). All data were recorded using continuous wave chemical exchange saturation transfer (CW-CEST) with a saturation power of 2 μT (t_{sat} = 2 s) for offsets from 1 to 4.5 ppm at 37°C. (C) the concentration dependence of the Cr CEST signal R_{guan}^{max}. The values were obtained from peak intensities of the guanidinium CEST *Z*-spectra (Z_{guan}^{ss}) and the measured R_1 values using equation 1. The broken line ($y = 0.0096x$) is drawn for visual guidance of the linear dependence range

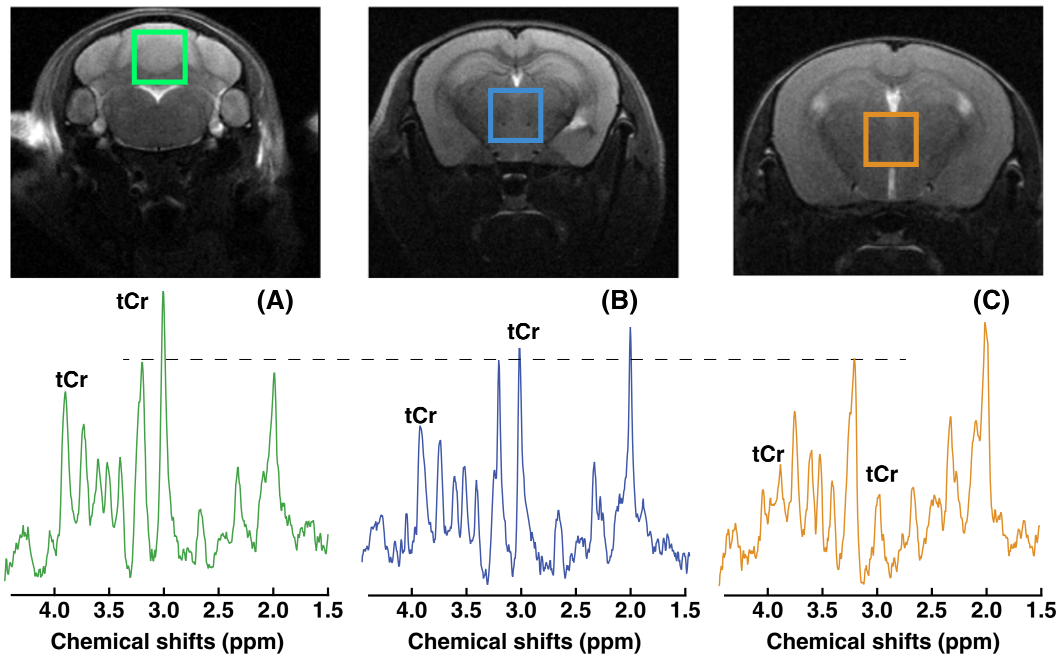

FIGURE 3 *In vivo* ^{1}H nuclear magnetic resonance (NMR) spectra measured from regions of the cerebellum in a wild-type (WT) mouse (A), the midbrain in a WT mouse (B) and the midbrain in a guanidinoacetate *N*-methyltransferase-deficient (GAMT−/−) mouse. The peak intensities were normalized to the choline (Cho) peak and a broken line was drawn for visual guidance of the relative peak intensities. tCr, total creatine. The corresponding T_2-weighted images of the mouse brain with a rapid acquisition with relaxation enhancement (RARE) sequence with the volumes of interest (VOIs) are plotted above the spectra. RARE magnetic resonance imaging (MRI): in-plane resolution, 100 × 100 μm^2; slice thickness, 1 mm

TABLE 1 LCModel-based tissue concentrations (mM) of brain metabolites with exchangeable protons at 1.95 ppm. Wild-type (WT), n = 3; guanidinoacetate *N*-methyltransferase deficiency (GAMT−/−), n = 3

	Cerebellum (WT)	Midbrain (WT)	Midbrain (GAMT−/−)
Glu + Gln	19.2 ± 1.3	19.5 ± 1.6	18.6 ± 0.4
GABA	3.3 ± 0.6	4.3 ± 0.6	3.4 ± 0.7
tCr	17.6 ± 1.9	13.7 ± 1.5	0.7 ± 0.6[a]

GABA, γ-aminobutyric acid; Gln, glutamine; Glu, glutamate; tCr, total creatine.

[a]Statistical significance p < 0.001.

Typical *Z*-spectra recorded in comparable VOIs are shown for WT and GAMT−/− mice in Figure 4A,B. The intensity of the broad background signal differed slightly between the two types of mouse, possibly because of the different MTC and DS values in the slightly mismatched brain locations. Therefore, *Z*-spectra were aligned on the intensity scale using the CEST signal from amide protons (3.5 ppm) by translation along the intensity axis. In principle, other offsets far away from 1.95 ppm can also be used for the alignment. The amide proton peak at 3.5 ppm was selected as this peak is helpful for the identification of not only the mismatch in intensity between two *Z*-spectra, but also any frequency shift cause by differences in B_0 inhomogeneity. The alignment is of high quality visually for the saturation power of 1 μT, as seen from the *Z*-spectra. However, the alignment was more difficult at high saturation power (2 μT), and a slight mismatch was found between WT and GAMT−/− mice for the 1–1.5-ppm range. Comparison between the two types of mouse shows some residual signal at 1.95 ppm in the GAMT−/− mouse, but we know from MRS experiments that there is minimal tCr remaining. This indicates that an appreciable portion of the guanidinium peak at 1.95 ppm is from a different source for the *Z*-spectra recorded at 1 μT, which, based on the eggwhite phantom results, can be tentatively assigned to mobile protein signals. This contribution should not change between mouse types, in line with the similar peak intensity found at the amide proton offset (3.5 ppm) for WT and GAMT−/− mice (Figure 4C). However, the protein contribution to the guanidinium peak at the higher saturation power of 2 μT is suppressed significantly, as shown in the *Z*-spectrum of the GAMT−/− mouse (Figure 4H), with tCr thus becoming the dominant component for the peak at 1.95 ppm in the WT mouse at such high saturation power (Figure 4F). The *Z*-spectra (1–3 ppm) of the brains of WT and GAMT−/− mice, together with the fitted *Z*-spectra using Equations 7, 8, 10 and 11, are plotted in Figure 4E–H. With a fourth-order polynomial function, the broad background *Z*-spectra can be well fitted between 1 and 3 ppm.

When multiple ST contributions are present, such as in brain tissues, observation and separation of the guanidinium peak as a function of the saturation power is complicated, because the observed CEST spectrum is not a simple zero-order superposition of all saturation contributions (Equation 8).[38] The tCr CEST contrast will be scaled down by the spillover effect from the strong background signals from DS and MTC (Equation 9). In order to determine the optimal saturation power for the *in vivo* CW-CEST experiments, a series of experiments with different saturation powers

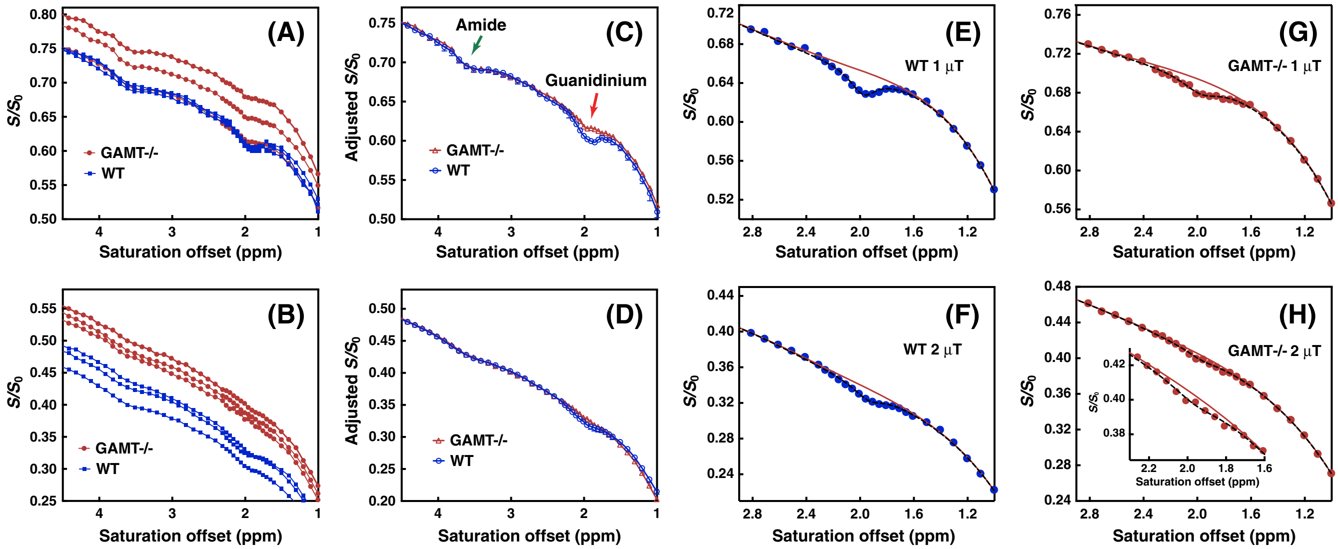

FIGURE 4 *Z*-spectra recorded in wild-type (n = 3) and guanidinoacetate *N*-methyltransferase-deficient (GAMT−/−) (n = 3) mice for the midbrain volume of interest (VOI) in Figure 3 using continuous wave chemical exchange saturation transfer (CW-CEST) with saturation power of 1 μT (A) and 2 μT (B). *Z*-spectra collected at 1 μT (C) and 2 μT (D) aligned with the *Z*-spectrum of one of the WT mice using the signal at 3.5 ppm as a reference for translation along the intensity scale. Fitted *Z*-spectra (1–3-ppm range, broken line) for the brain of WT (E, F) and GAMT−/− (G, H) mice with saturation power of 1 μT (E, G) and 2 μT (F, H), respectively. The inset in Figure 4H shows the fitting of the small residual. The full curves are the polynomial fits of the $Z^{ss}_{background}$ signal around 1.95 ppm

were performed on both WT and GAMT−/− mouse brains. Typical *Z*-spectra of a WT mouse brain as a function of saturation power are plotted in Figure 5A, and the observed guanidinium CEST signal difference (ΔZ) and calculated R^{max}_{guan} results are presented in both Table 2 and Figure 5B,C. An average relaxation time of R_1 = 0.55 ± 0.03 s^{-1} for mouse brain was used to calculate R^{max}_{guan}. The guanidinium CEST signal difference (ΔZ) for the WT mouse, which is defined by the difference between the guanidinium maximum signal and the background signal at 1.95 ppm (Figure 1), increased slightly with saturation power before reaching a maximum around 1.2 μT, and dropping rapidly at higher powers (Figure 5B), presumably because of the spillover effect prohibiting accurate quantification. The observed guanidinium CEST signal (ΔZ) was significantly lower for the brains of GAMT −/− mice, and the values predominantly decreased as a function of saturation power. This apparent drop is again caused by the dominance of the broad background, similar to the WT mice at higher power. The results in Figure 4 clearly show that the component of the guanidinium peak for GAMT−/− mice is much smaller than that of the WT mouse brain. The guanidinium CEST signal peaks obtained by fitting the *Z*-spectra using Equations 7, 8, 10 and 11 are presented in Figure 5C. After spillover correction, this fitted R^{max}_{guan} shows a strong increase with respect to the saturation power, which is expected because of the higher labeling efficiency.

The fitted R^{max}_{guan} can be correlated with the Cr and PCr concentrations from MRS by the relationship:

$$R^{max}_{guan} = R_{protein} + r_{Cr} \cdot [\mathrm{Cr}] + r_{PCr} \cdot [\mathrm{PCr}] \tag{12}$$

where $R_{protein}$ is the protein signal in R_{guan} (intercept of the curve in Figure 5D), and r_{Cr} and r_{PCr} are the apparent relaxivites that must be multiplied with the concentrations of Cr and PCr in brain (slope of the curve in Figure 5D) to obtain the apparent relaxation contributions. Assuming the PCr to Cr concentration ratio to be constant, Equation 12 can be simplified to:

$$R^{max}_{guan} = R_{protein} + r_{tCr} \cdot [tCr] \tag{13}$$

Using the data in Figure 5D, $R_{protein}$ = 0.015 s^{-1} and r_{tCr} = 1.28 × 10^{-3} $s^{-1}mM^{-1}$ were determined for a saturation power of 1 μT, whereas $R_{protein}$ = 0.016 s^{-1} and r_{tCr} = 3.42 × 10^{-3} $s^{-1}mM^{-1}$ were obtained for a saturation power of 2 μT. The value of $R_{protein}$ is determined by the protein concentration and saturation power.

We calculated tCr maps for the cerebellum of the WT mouse brain using Equations 7, 8, 10, 11 and 13, which are shown in Figure 6. The tCr concentration in the cerebellum region is about 18.7 ± 4mM, whereas the tCr concentration is much lower for the medulla (14.1 ± 3mM). The tCr concentration measured by CEST is slightly higher than the values, 15.5–16.8mM, determined by the MRS technique.[54,55] The corresponding *C*0 maps are also presented. As discussed in Equation 11, the *C*0 map is a combined effect mainly from DS and MTC plus all other ST processes.

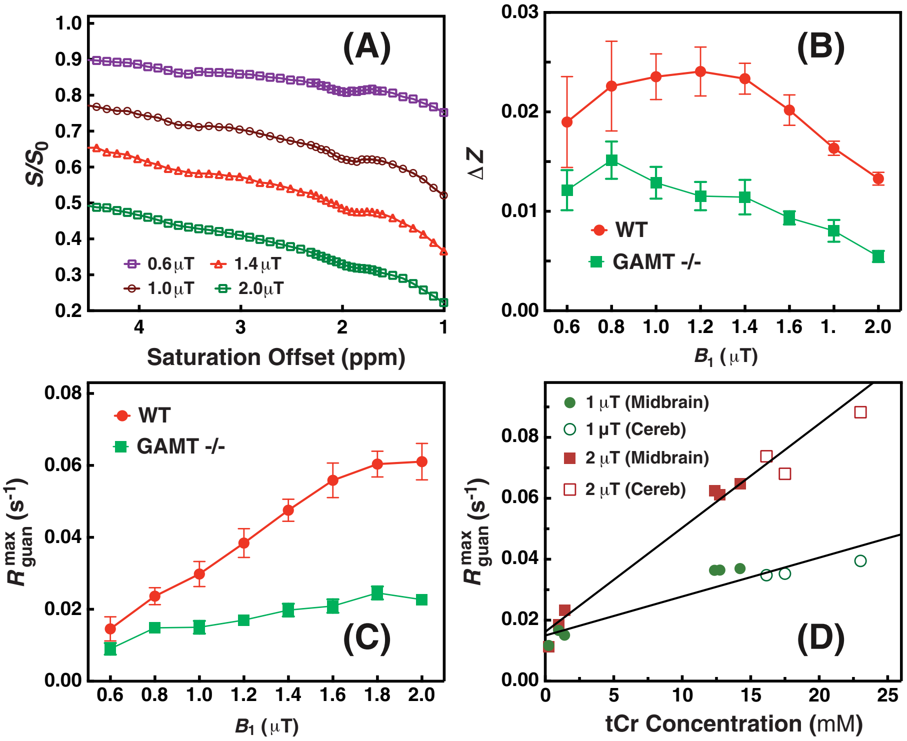

FIGURE 5 (A) Typical *Z*-spectra of wild-type (WT) mouse brain for the midbrain volume of interest (VOI) in Figure 3 recorded with different saturation powers (0.6, 1, 1.4 and 2 μT). (B) The observed guanidinium chemical exchange saturation transfer (CEST) difference signal Δ*Z* and (C) the guanidinium CEST signal (R_{guan}^{max}) with respect to the saturation power for WT and guanidinoacetate *N*-methyltransferase-deficient (GAMT−/−) mice. The error bar represents the standard deviation over the midbrain region. (D) The fitted peak intensities of the guanidinium CEST signals from the midbrain (filled circles and squares) and cerebellum (open circles and squares) of WT mice (*n* = 3) and from the midbrain region of GAMT−/− mice (*n* = 3) with their corresponding total creatine (tCr) concentrations quantified relative to magnetic resonance spectroscopy (MRS). Lines are the linear least-squares fitted curves, leading to $R_{protein}$ = 0.015 s^{-1} and r_{tCr} = 1.28 × 10^{-3} $s^{-1}mM^{-1}$ (R^2 = 0.87) for 1 μT and $R_{protein}$ = 0.016 s^{-1} and r_{tCr} = 3.41 × 10^{-3} $s^{-1}mM^{-1}$ (R^2 = 0.94) for 2 μT

TABLE 2 $Z_{background}^{ss}$ and *Z* at 1.95 ppm for the midbrain of guanidinoacetate *N*-methyltransferase-deficient (GAMT−/−) and wild-type (WT) mice measured with different saturation powers. The calculated R_{guan}^{max} values using equations 7 and 8 are also listed

	WT			GAMT−/−		
Power (μT)	$Z_{background}^{ss}$	Z (10^{-2})	R_{guan}^{max} (10^{-2})	$Z_{background}^{ss}$	Z (10^{-2})	R_{guan}^{max} (10^{-2})
0.6	0.804 ± 0.02	1.9 ± 0.5	1.6 ± 0.4	0.821 ± 0.02	1.2 ± 0.2	1.0 ± 0.2
0.8	0.718 ± 0.02	2.3 ± 0.5	2.4 ± 0.4	0.736 ± 0.02	1.5 ± 0.2	1.6 ± 0.2
1.0	0.659 ± 0.02	2.4 ± 0.2	3.0 ± 0.4	0.665 ± 0.02	1.3 ± 0.2	1.6 ± 0.2
1.2	0.608 ± 0.03	2.4 ± 0.3	3.6 ± 0.4	0.569 ± 0.03	1.1 ± 0.1	1.7 ± 0.1
1.4	0.536 ± 0.03	2.3 ± 0.2	4.5 ± 0.3	0.505 ± 0.03	1.1 ± 0.2	2.2 ± 0.2
1.6	0.473 ± 0.04	2.0 ± 0.2	5.0 ± 0.5	0.435 ± 0.04	0.9 ± 0.1	2.3 ± 0.2
1.8	0.409 ± 0.04	1.6 ± 0.1	5.4 ± 0.4	0.391 ± 0.04	0.8 ± 0.1	2.6 ± 0.2
2.0	0.364 ± 0.04	1.3 ± 0.1	5.5 ± 0.4	0.351 ± 0.04	0.5 ± 0.1	2.3 ± 0.2

4 | DISCUSSION

Based on the comparison of brain *Z*-spectra recorded in the brains of WT and GAMT−/− mice, this study provides strong evidence that the peak of tCr is the major contributor to the well-defined guanidinium peak at 1.95 ppm in the *Z*-spectrum. However, there is a significant part of the guanidinium peak from another source, which we attribute to mobile proteins in line with a similar resonance found in eggwhite phantoms. The ratio between the protein and tCr peak contributions varies with different experimental conditions. The tCr contribution from PCr is expected to be small compared with Cr because of the much slower exchange rate, but could still contribute 17 ± 5% of the tCr signal at a saturation power of 2 μT, based on our phantom studies under physiological conditions for brain.

The guanidinium peak does not completely arise from tCr, as shown by a small residual guanidinium peak in the *Z*-spectra from GAMT−/− mouse brains (Figure 4C) and the non-zero intercept of the correlation analysis between MRS and the guanidinium peak (Figure 5D). The residual

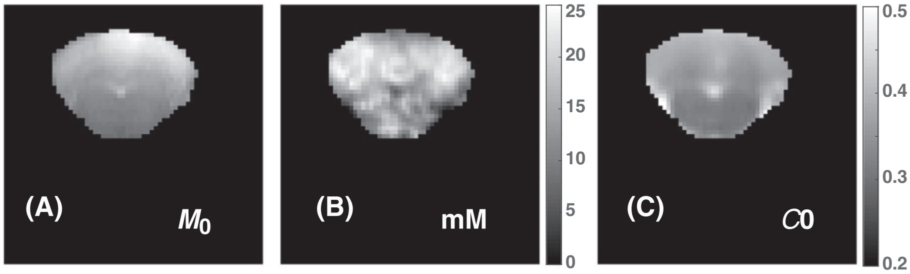

FIGURE 6 (A) M_0 images, (B) total creatine (tCr) maps and (C) *C*0 maps [the zero-order coefficient for the polynomial function to fit the chemical exchange saturation transfer (CEST) background] for the cerebellum of a wild-type (WT) mouse. The maps were calculated pixel by pixel by fitting the *Z*-spectra between 1 and 3 ppm, recorded using continuous wave (CW)-CEST with a saturation power of 2 μT. *C*0 maps contain contributions mainly from direct water saturation (DS) and magnetization transfer contrast (MTC), and clearly show strong white matter and gray matter contrast, similar to MTC maps

may contain signal from the guanidinium, amine or aromatic protons of mobile proteins as confirmed by the *Z*-spectra of the eggwhite phantom (Figure 2A) and a previous study on tumors and tissue homogenates.[12,42] The relative contribution of the tCr signal to the guanidinium peak depends on the saturation power. At a saturation power of 1 μT (for the current saturation time of 2 s), the tCr signal was derived to be 65% of the total signal. The tCr contribution to the guanidinium peak increased to 80 ± 7% with a saturation power of 2 μT, in which Cr contributed 83 ± 5% and PCr contributed 17 ± 5%. Thus, the guanidinium peak of WT mice is dominated by the tCr signal at a saturation power of 2 μT. Therefore, a saturation power of 2 μT is suggested for future studies to obtain a relatively clean (80%) tCr signal. The current study suggests that the detection of Cr in the brain must take into account the contributions from mobile proteins. During the time of revision of the current paper, a comparison study on the tissue homogenates of rat brain was published, which suggested that protein contributes about 34% to the peak at 1.95 ppm (1 μT and 5 s CW-CEST), which perfectly matches with our finding at 1 μT (35%).[42] This study thus verifies our quantification method and results. Our results also show that the CEST signal of the guanidinium protons in the protein shows much less of an increase as a function of saturation power than that of tCr, as judged from Figure 5C. This suggests that the exchange rates of the guanidinium protons in the proteins are much lower than those in Cr. This property is helpful for us to design methods to suppress the protein contamination in tCr mapping. The contributions of mobile proteins and tCr to the CEST signal depend on pH, temperature and the saturation parameters used as discussed above. Therefore, the calibration of tCr by MRS using Equation 13 may not be valid when the ratio of PCr and Cr changes during energy use, such as for brain activity or exercise in muscle.

The extraction of the tCr signal in the current study provides an opportunity to estimate the contribution of tCr in the conventional asymmetry analysis method, i.e. $\mathrm{MTR}_{asym} = (S_{-2ppm} - S_{2ppm})/S_0$, where $S_{\pm 2ppm}$ are the saturated signal intensities at ±2 ppm.[27,31,35] The CEST contrast due to tCr at 1.95 ppm is measured to be about 1.3 ± 0.1% of the water signal in the WT mouse brain (2 μT and 2 s length, Figure 5B). The majority of the saturation effects at 1.95 ppm (about 77% of the signal) derives from DS, aromatic protons and amine/guanidinium protons of proteins, and MTC. When performing an asymmetry analysis on the CEST *Z*-spectra, MTR_{asym} was found to be 5.2 ± 0.5% (2 μT and 2 s length), whereas the tCr signal was not reduced by the asymmetry analysis. Therefore, conventional MTR_{asym} is an efficient way to remove DS and the symmetric part of MTC, which constitute the majority of the contaminations to the tCr signal in brain, but relatively large contributions still exist on the order of the magnitude of the tCr signal.

Compared with the conventional Lorentzian line-shape fitting method or asymmetry analysis method, the current simple approach not only provides a cleaner tCr mapping, but also shortens the total experimental time by acquiring a small portion of the full *Z*-spectra (1–4.5 ppm). When acquiring the concentration map of tCr, a B_0 map on mouse brain is not necessary as the concentration was obtained from the extracted guanidinium peak at 1.95 ppm (R_{guan}^{max}) based on the maximum of the peak fit on the observed guanidinium CEST signal (ΔZ). Although the calibration method was demonstrated only for brain, it is expected that it could also be used to estimate and map tCr in muscle.[38] In skeletal muscle, the concentrations of Cr and PCr are on the order of 7mM[66] and 20–30mM,[67] respectively, and thus tCr is significantly higher than in brain. Furthermore, MTC is close to symmetric in muscle, which is different from the line-shape of MTC in the brain.[68] Therefore, the conventional MTR_{asym} method may be a simple and valid approximation to suppress the majority of the DS and MTC contaminations. However, there is a large signal contribution from the OH groups of glycogen and it is unclear how much signal from mobile protein would be there. However, the exchange rate of OH is much higher and it may be possible to separate guanidinium protons and OH protons using rate-based editing approaches.

The current approach is ready to be transferred to high-field clinical scanners, such as 7-T scanners. When the magnetic field is reduced to 3 T, the exchange rate of Cr can no longer be considered to be in the intermediate exchange range. The guanidinium peak of Cr coalesces with water peaks because of the reduced shift difference at these lower fields, and overlaps with many other exchanging protons. Then, the separation of the Cr signal will be very challenging, but the extraction of PCr may still be feasible because of its much slower exchange rate.[25]

5 | CONCLUSION

In this work, a peak-fitting scheme assuming a two-pool model was used to estimate the Cr/PCr contributions to the CEST *Z*-spectra of mouse brain at 11.7 T. CEST spectra from WT and GAMT−/− mouse brains revealed that the guanidinium peak at 1.95 ppm arises from both tCr and a residual, assigned to mobile proteins. The ratio of these contributions depends on the saturation power. Quantitative tCr maps may be generated using the fitted peak of the guanidinium signal in the CEST spectrum after calibration using *in vivo* MRS.

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